Energy is probably the most important value when generating caustics. The intensity of caustics pretty much follows the energy of the photons. When the value is too low, the caustics appear faint, but high numbers can easily blow out a render. The position of the light is another important value to consider. By default, photon intensity decays at the square of the distance from the light. This means that placing the light twice as far away reduces the intensity of the photons by a factor of four.

Conversely, moving the light twice as close quadruples the intensity. Finally, the Decay attribute also plays a factor in how the caustic is generated. This number represents the exponential value at which the light energy decays. Generally, this is kept at the default of 2, which simulates the inverse square law and real-world lighting, but lower numbers can extend the power of the light, and higher numbers will limit it.

5.7 Render Elements

The Render Elements tab of the Render Scene window lets you separate out different parts of the render and save them into individual image files. This can be useful when you composite images or work with image-processing or special-effects software. Rendering such elements as shadows or specularity to separate files allows you to adjust their levels inside a compositing package to achieve a more-seamless integration of 3D with live action or other rendered elements.

5.8 Backburner

Backburner is an excellent batch and network rendering utility for 3ds Max. It allows you to set up a render farm of multiple machines and manage multiple renders. Backburner is a separate application from 3ds Max and must be started from the Windows Start menu. Backburner has three main programs: manager, server, and monitor. These are located in the Start menu, under Autodesk/Backburner.

When rendering within 3ds Max, Backburner can be accessed by toggling the Net Render box in the render output section of the Render Scene window. This brings up a window where you can assign the render job to specific servers.

Network rendering is started in three steps:

1. The network manager program is started on one machine.
2. The render servers are started on the rest of the machines.
3. Jobs are submitted.